

CS545 - Cloud Native Platform Engineering

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Lab 3

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Part 1

D1.

```
((base) anthilyra@Anthis-MacBook-Air lab-p1 % pulumi up
Previewing update (dev):
  Type                                Name                                Plan
+   pulumi:pulumi:Stack                novaspark-vpc-dev                  create
+   -- aws:ec2:Eip                      nat-eip                           create
+   -- aws:ec2:Vpc                      novaspark-vpc                     create
+   -- aws:ec2:Subnet                  public-subnet-b                   create
+   -- aws:ec2:Subnet                  private-subnet-b                  create
+   -- aws:ec2:RouteTable              private-rt                         create
+   -- aws:ec2:SecurityGroup           bastion-sg                        create
+   -- aws:ec2:Subnet                  public-subnet-a                   create
+   -- aws:ec2:RouteTable              public-rt                         create
+   -- aws:ec2:RouteTableAssociation    public-rt-assoc-a                 create
+   -- aws:ec2:RouteTableAssociation    private-rt-assoc-a                create
+   -- aws:ec2:Subnet                  private-subnet-a                  create
+   -- aws:ec2:SecurityGroup           private-sg                        create
+   -- aws:ec2:InternetGateway          novaspark-igw                     create
+   -- aws:ec2:RouteTableAssociation    public-rt-assoc-b                 create
+   -- aws:ec2:Route                  public-default-route              create
+   -- aws:ec2:RouteTableAssociation    private-rt-assoc-b                create
+   -- aws:ec2:NatGateway              novaspark-nat                     create
+   -- aws:ec2:Route                  private-default-route              create
+   -- aws:ec2:Instance                private-instance                  create
+   -- aws:ec2:Instance                bastion                           create
```

```
Outputs:
bastionPublicDns : [unknown]
bastionPublicIp   : [unknown]
natEip            : [unknown]
privateInstanceIp : [unknown]
privateSubnetA    : [unknown]
privateSubnetB    : [unknown]
publicSubnetA     : [unknown]
publicSubnetB     : [unknown]
sshCommand        : [unknown]
sshHopCommand     : [unknown]
vpcCidr           : "10.0.0.0/16"
vpcId             : [unknown]
```

```
Resources:
+ 21 to create
```

Do you want to perform this update? yes

```
Updating (dev):
  Type                                Name                                Status
+   pulumi:pulumi:Stack                novaspark-vpc-dev                  created (130s)
+   -- aws:ec2:Vpc                      novaspark-vpc                     created (2s)
+   -- aws:ec2:Eip                      nat-eip                           created (2s)
+   -- aws:ec2:Subnet                  public-subnet-b                   created (11s)
+   -- aws:ec2:Subnet                  private-subnet-b                  created (0.87s)
+   -- aws:ec2:RouteTable              private-rt                         created (0.87s)
+   -- aws:ec2:Subnet                  public-subnet-a                   created (11s)
+   -- aws:ec2:SecurityGroup           bastion-sg                        created (3s)
+   -- aws:ec2:SecurityGroup           private-sg                        created (3s)
+   -- aws:ec2:Subnet                  private-subnet-a                  created (0.87s)
+   -- aws:ec2:InternetGateway          novaspark-igw                     created (0.92s)
+   -- aws:ec2:RouteTable              private-rt                         created (0.87s)
+   -- aws:ec2:RouteTableAssociation    private-rt-assoc-a                 created (0.47s)
+   -- aws:ec2:RouteTableAssociation    private-rt-assoc-b                created (0.56s)
+   -- aws:ec2:Route                  public-default-route              created (0.82s)
+   -- aws:ec2:Instance                private-instance                  created (13s)
+   -- aws:ec2:RouteTableAssociation    private-rt-assoc-a                created (0.44s)
+   -- aws:ec2:Instance                bastion                           created (13s)
+   -- aws:ec2:NatGateway              novaspark-nat                     created (114s)
```

```
+   -- aws:ec2:Instance                bastion                           created (13s)
+   -- aws:ec2:NatGateway              novaspark-nat                     created (114s)
+   -- aws:ec2:RouteTableAssociation    public-rt-assoc-b                 created (0.41s)
+   -- aws:ec2:Route                  private-default-route              created (0.87s)
```

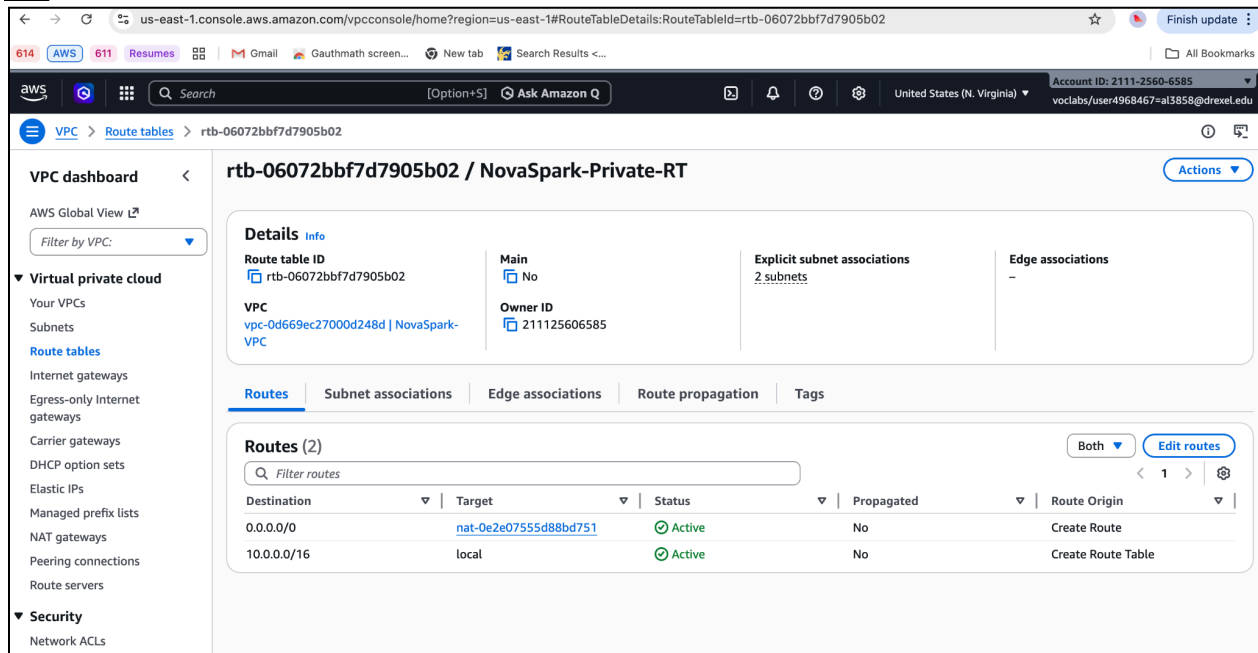
```
Outputs:
bastionPublicDns : "ec2-100-53-4-167.compute-1.amazonaws.com"
bastionPublicIp   : "100.53.4.167"
natEip            : "34.239.47.22"
privateInstanceIp : "10.0.11.5"
privateSubnetA    : "subnet-0e2635a8ca96a4c27"
privateSubnetB    : "subnet-0ba1f6f566aa8a8f1"
publicSubnetA     : "subnet-0eb0399f250b010197"
publicSubnetB     : "subnet-05d0dbaebc273a155"
sshCommand        : "ssh -A -i ~/.ssh/labuser.pem ec2-user@ec2-100-53-4-167.compute-1.amazonaws.com"
sshHopCommand     : "ssh ec2-user@10.0.11.5 # run this FROM the bastion, after SSH-ing in with -A"
vpcCidr           : "10.0.0.0/16"
vpcId             : "vpc-0d669ec27000d248d"
```

```
Resources:
+ 21 created
```

Duration: 2m14s

(base) anthilyra@Anthis-MacBook-Air lab-p1 %

D2.



The screenshot displays the AWS Management Console interface for a Route Table. The breadcrumb navigation shows the path: VPC > Route tables > rtb-06072bbf7d7905b02. The main heading is "rtb-06072bbf7d7905b02 / NovaSpark-Private-RT".

Details:

- Route table ID: rtb-06072bbf7d7905b02
- VPC: vpc-0d669ec27000d248d | NovaSpark-VPC
- Main: No
- Owner ID: 211125606585
- Explicit subnet associations: 2 subnets
- Edge associations: -

Routes (2)

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-0e2e07555d88bd751	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

D3.

An Internet Gateway is the VPC's connection to the internet and is used by the public subnets. Instances in public subnets have public IP addresses and can send and receive traffic. However, a NAT Gateway is more like a one-way door for private subnets, where instances can send only outbound requests to the internet, but the internet cannot start connections back to them. Private instances can not use the Internet Gateway directly because they do not have public IPs and their route table sends traffic though the NAT instead.

D4.

The bastion host pattern exists because private instances cannot be accessed directly from the internet so the bastion acts as an SSH entry point inside the public subnet. First we SSH into the bastion host and from there we can SSH into the private instances. This solves the problem of accessing private servers without giving them public access. However, the bastion is a security risk because port 22 is open to the internet, so anyone on the internet could try to access it. One improvement would be restricting SSH access to specific IP addresses or using AWS Systems Manager Session Manager instead of SSH.

D5.

```
((base) anthilyra@Anthis-MacBook-Air lab-p1 % pulumi destroy
Previewing destroy (dev):
Type                                     Name                                     Plan
- ┬─ pulumi:pulumi:Stack                 novaspark-vpc-dev                     delete
- │ ┬─ pulumi:pulumi:Stack                 novaspark-vpc-dev                     delete
- │ │ ┬─ aws:ec2:RouteTableAssociation     private-rt-assoc-a                   delete
- │ │ │ ┬─ aws:ec2:RouteTable              public-rt                             delete
- │ │ │ │ ┬─ aws:ec2:Subnet                private-subnet-a                     delete
- │ │ │ │ │ ┬─ aws:ec2:RouteTableAssociation public-rt-assoc-b                     delete
- │ │ │ │ │ │ ┬─ aws:ec2:Route              public-default-route                 delete
- │ │ │ │ │ │ │ ┬─ aws:ec2:Instance         bastion                              delete
- │ │ │ │ │ │ │ │ ┬─ aws:ec2:Subnet          public-subnet-b                     delete
- │ │ │ │ │ │ │ │ │ ┬─ aws:ec2:SecurityGroup bastion-sg                           delete
- │ │ │ │ │ │ │ │ │ │ ┬─ aws:ec2:Route      private-default-route                 delete
- │ │ │ │ │ │ │ │ │ │ │ ┬─ aws:ec2:RouteTableAssociation public-rt-assoc-a                   delete
- │ │ │ │ │ │ │ │ │ │ │ │ ┬─ aws:ec2:Subnet                private-subnet-a                     delete
- │ │ │ │ │ │ │ │ │ │ │ │ │ ┬─ aws:ec2:RouteTableAssociation private-rt-assoc-b                   delete
- │ │ │ │ │ │ │ │ │ │ │ │ │ │ ┬─ aws:ec2:NatGateway          novaspark-nat                       delete
- │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ ┬─ aws:ec2:Instance         private-instance                     delete
- │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ ┬─ aws:ec2:InternetGateway novaspark-igw                       delete
- │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ ┬─ aws:ec2:RouteTable      private-rt                           delete
- │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ ┬─ aws:ec2:Subnet          private-subnet-b                     delete
- │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ ┬─ aws:ec2:Eip             nat-eip                             delete
- │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ ┬─ aws:ec2:SecurityGroup private-sg                           delete
- │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ ┬─ aws:ec2:Vpc             novaspark-vpc                       delete

Resources:
- 21 to delete

Do you want to perform this destroy? yes
Destroying (dev):
Type                                     Name                                     Status
- ┬─ pulumi:pulumi:Stack                 novaspark-vpc-dev                     deleted (0.00s)
- │ ┬─ pulumi:pulumi:Stack                 novaspark-vpc-dev                     deleted (1s)
- │ │ ┬─ aws:ec2:Route                    private-default-route                 deleted (1s)
- │ │ │ ┬─ aws:ec2:Instance                bastion                              deleted (30s)
- │ │ │ │ ┬─ aws:ec2:Instance              private-instance                     deleted (20s)
- │ │ │ │ │ ┬─ aws:ec2:RouteTableAssociation public-rt-assoc-a                     deleted (0.41s)
- │ │ │ │ │ │ ┬─ aws:ec2:Route              public-default-route                 deleted (0.71s)
- │ │ │ │ │ │ │ ┬─ aws:ec2:RouteTableAssociation private-rt-assoc-a                   deleted (0.52s)
- │ │ │ │ │ │ │ │ ┬─ aws:ec2:RouteTableAssociation public-rt-assoc-b                   deleted (0.49s)
- │ │ │ │ │ │ │ │ │ ┬─ aws:ec2:RouteTableAssociation private-rt-assoc-b                   deleted (0.50s)
- │ │ │ │ │ │ │ │ │ │ ┬─ aws:ec2:NatGateway          novaspark-nat                       deleted (71s)
- │ │ │ │ │ │ │ │ │ │ │ ┬─ aws:ec2:Subnet          private-subnet-a                     deleted (0.70s)
- │ │ │ │ │ │ │ │ │ │ │ │ ┬─ aws:ec2:InternetGateway novaspark-igw                       deleted (0.72s)
- │ │ │ │ │ │ │ │ │ │ │ │ │ ┬─ aws:ec2:Subnet          private-subnet-b                     deleted (0.76s)
- │ │ │ │ │ │ │ │ │ │ │ │ │ │ ┬─ aws:ec2:SecurityGroup bastion-sg                           deleted (0.86s)
- │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ ┬─ aws:ec2:RouteTable      private-rt                           deleted (0.81s)
- │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ ┬─ aws:ec2:Subnet          public-subnet-a                     deleted (0.86s)
- │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ ┬─ aws:ec2:SecurityGroup private-sg                           deleted (0.84s)
- │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ ┬─ aws:ec2:RouteTable      public-rt                           deleted (0.82s)
- │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ ┬─ aws:ec2:Subnet          public-subnet-b                     deleted (0.81s)
- │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ ┬─ aws:ec2:Eip             nat-eip                             deleted (0.45s)
- │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ ┬─ aws:ec2:Vpc             novaspark-vpc                       deleted (0.70s)

Resources:
- 21 deleted

Duration: 1m17s
```

Part 2

D6.

```
or visit https://pulumi.com/docs/install/ for manual instructions and release notes.
(base) anthilyra@Anthis-MacBook-Air lab-p1 % eval $(ssh-agent -s)
ssh-add ~/.ssh/labsuser.pem
Agent pid 96081
Identity added: /Users/anthilyra/.ssh/labsuser.pem (/Users/anthilyra/.ssh/labsuser.pem)
(base) anthilyra@Anthis-MacBook-Air lab-p1 % ssh-add -l
2048 SHA256:T9TgdRkK/KBBtALwCgkftd0Ebi0igca7aIT3F9JLSqs /Users/anthilyra/.ssh/labsuser.pem (RSA)
(base) anthilyra@Anthis-MacBook-Air lab-p1 %
```

D7.

```
Outputs:
bastionPublicDns : "ec2-98-93-165-111.compute-1.amazonaws.com"
bastionPublicIp   : "98.93.165.111"
natEip            : "3.218.133.123"
privateInstanceId : "i0-0.11.247"
privateSubnetA    : "subnet-07c7b6d2ded8ce208c"
privateSubnetB    : "subnet-0521ffffbf790d7a99"
publicSubnetA     : "subnet-0376ecf2a57bd6add"
publicSubnetB     : "subnet-01b9142180f8c91ee"
sshCommand        : "ssh -A -i ~/.ssh/labsuser.pem ec2-user@ec2-98-93-165-111.compute-1.amazonaws.com"
sshHopCommand     : "ssh ec2-user@10.0.11.247 # run this FROM the bastion, after SSH-ing in with -A"
vpcCidr           : "10.0.0.0/16"
vpcId             : "vpc-0bbe18275f67b3c96"
```

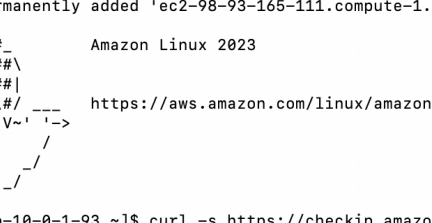
Resources:

- + 21 created

Duration: 2m17s

warning: A new version of Pulumi is available. To upgrade from version '3.232.0' to '3.235.0', run
\$ curl --ssl https://get.pulumi.com | sh
or visit https://pulumi.com/docs/install/ for manual instructions and release notes.

```
(base) anthilyra@Anthis-MacBook-Air lab-p1 % eval $(ssh-agent -s)
ssh-add ~/.ssh/labsuser.pem
Agent pid 96081
Identity added: /Users/anthilyra/.ssh/labsuser.pem (/Users/anthilyra/.ssh/labsuser.pem)
(base) anthilyra@Anthis-MacBook-Air lab-p1 % ssh-add -l
2048 SHA256:T9TgdRRKk/KBBtAlwCgkfTdOEbi0igca7aIT3F9JLSqs /Users/anthilyra/.ssh/labsuser.pem (RSA)
(base) anthilyra@Anthis-MacBook-Air lab-p1 % ssh -A -i ~/.ssh/labsuser.pem ec2-user@ec2-98-93-165-111.compute-1.amazonaws.com
The authenticity of host 'ec2-98-93-165-111.compute-1.amazonaws.com (98.93.165.111)' can't be established.
ED25519 key fingerprint is SHA256:vgtTZ6a7XB7/zapOHVGAuAu06HYpnckKu2Y3pqZbr5pY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-98-93-165-111.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
```



```
.#_
~\_ ##### Amazon Linux 2023
~~\_\_####\
~~ \###|
~~ \#/ ----- https://aws.amazon.com/linux/amazon-linux-2023
~~ V~' '->
~~~~
~~-. _/_/
~~_/'_/
~/m/'
```

```
[ec2-user@ip-10-0-1-93 ~]$ curl -s https://checkip.amazonaws.com
98.93.165.111
[ec2-user@ip-10-0-1-93 ~]$
```

D8.

```
~~~\#/ --- https://aws.amazon.com/linux/amazon-linux-2023
~~~V~' '->
~~~~/
~~~._./ -/_/
~/m/'
```

[ec2-user@ip-10-0-11-247 ~]\$ hostname -I
10.0.11.247
[ec2-user@ip-10-0-11-247 ~]\$

D9

```
10.0.11.247
[ec2-user@ip-10-0-11-247 ~]$ curl -sf --max-time 3 http://169.254.169.254/latest/meta-data/public-ipv4 || echo "No public IP"
No public IP
[ec2-user@ip-10-0-11-247 ~]$
```


D10.

```
[ec2-user@ip-10-0-11-247 ~]$ curl -s https://checkip.amazonaws.com
3.218.133.123
[ec2-user@ip-10-0-11-247 ~]$
```

```
bastionPublicIp  : "98.93.165.111"
natEip           : "3.218.133.123"
privateInstanceIp: "10.0.11.247"
```

The returned IP matched the natEip Pulumi output (3.218.133.123). This happened because the private instance sends outbound internet traffic through the NAT Gateway, since it does not have its own public IP address. In larger environments, the IP may not match if there are multiple NAT Gateways or maybe different outbound routing configurations.

D11.

```
Resources:
  - 21 to delete

Do you want to perform this destroy? yes
Destroying (dev):
  Type                                     Name                                     Status
  - ┌ pulumi:pulumi:Stack                  novaspark-vpc-dev
  - │   ┌ pulumi:pulumi:Stack              novaspark-vpc-dev    deleted (0.01s)
  - │ │   ┌ aws:ec2:Route                  private-default-route deleted (1s)
  - │ │ │   ┌ aws:ec2:Instance              private-instance     deleted (30s)
  - │ │ │ │   ┌ aws:ec2:RouteTableAssociation public-rt-assoc-b    deleted (0.68s)
  - │ │ │ │ │   ┌ aws:ec2:RouteTableAssociation private-rt-assoc-a   deleted (0.75s)
  - │ │ │ │ │ │   ┌ aws:ec2:RouteTableAssociation public-rt-assoc-a   deleted (0.76s)
  - │ │ │ │ │ │ │   ┌ aws:ec2:Instance      bastion            deleted (30s)
  - │ │ │ │ │ │ │ │   ┌ aws:ec2:RouteTableAssociation private-rt-assoc-b   deleted (0.68s)
  - │ │ │ │ │ │ │ │ │   ┌ aws:ec2:Route      public-default-route   deleted (1s)
  - │ │ │ │ │ │ │ │ │ │   ┌ aws:ec2:NatGateway novaspark-nat        deleted (70s)
  - │ │ │ │ │ │ │ │ │ │ │   ┌ aws:ec2:SecurityGroup bastion-sg           deleted (0.83s)
  - │ │ │ │ │ │ │ │ │ │ │ │   ┌ aws:ec2:Subnet private-subnet-b    deleted (0.76s)
  - │ │ │ │ │ │ │ │ │ │ │ │ │   ┌ aws:ec2:Subnet private-subnet-a    deleted (0.63s)
  - │ │ │ │ │ │ │ │ │ │ │ │ │ │   ┌ aws:ec2:InternetGateway novaspark-igw        deleted (0.61s)
  - │ │ │ │ │ │ │ │ │ │ │ │ │ │ │   ┌ aws:ec2:Subnet public-subnet-b    deleted (0.72s)
  - │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │   ┌ aws:ec2:SecurityGroup private-sg           deleted (0.83s)
  - │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │   ┌ aws:ec2:RouteTable private-rt            deleted (0.72s)
  - │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │   ┌ aws:ec2:Subnet public-subnet-a    deleted (0.78s)
  - │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │   ┌ aws:ec2:RouteTable public-rt            deleted (0.70s)
  - │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │   ┌ aws:ec2:Eip nat-eip              deleted (0.36s)
  - │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │ │   ┌ aws:ec2:Vpc novaspark-vpc        deleted (0.64s)

Resources:
  - 21 deleted

Duration: 1m17s

The resources in the stack have been deleted, but the history and configuration associated with the stack are still maintained.
If you want to remove the stack completely, run `pulumi stack rm dev`.
(base) anthilyra@Anthis-MacBook-Air lab-p1 %
```

Context Paragraph Prompt

The reading explained that route tables control how traffic moves inside a VPC by deciding where network traffic is sent. In Lab 3 Part 2, I saw this when the public subnet used a route table connected to the Internet Gateway while the private subnet used a route table connected to the NAT Gateway instead. Because of this, I could SSH directly into the bastion host in the public subnet, but I could not SSH directly into the private instance from my laptop because it had no direct route to the internet. I first had to connect to the bastion host and then SSH into the private instance. The lab helped me better understand how routing actually works because I could directly see the difference between the public and private subnet connections. The reading explains how route tables work, but the lab showed how one routing change completely affects whether a server is publicly reachable or private while still having outbound internet access.